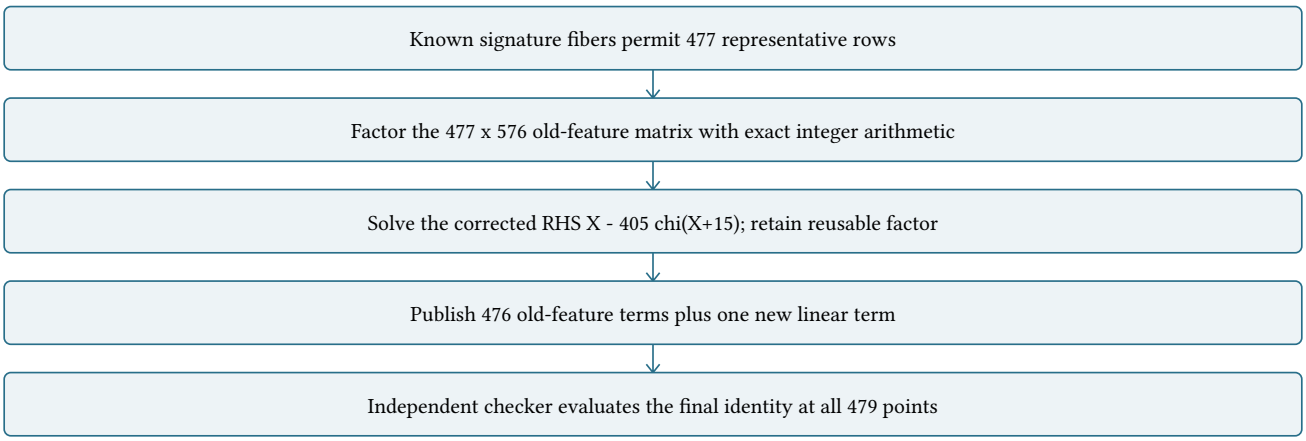
Explicit corrected-span representation

2026-11-03 · Exact identity certificate · No performance comparison

Constructed. The missing old-feature coefficients are now explicit. Over F479, X equals a cubic polynomial in characters at offsets0..14 plus 405chi(X+15). The program has476 nonzero old-feature terms and one new linear term. A checker independent of the solver evaluates the identity at all479 points using square-set characters.



Correctness gate. The two omitted rows173 and440 are known signature mates of retained rows25 and292. The correction makes the target constant within each pair. A fit on representatives alone is insufficient: the negative control fits uncorrected X on all477 representatives but fails globally at exactly173 and440. The repaired program passes every point.

Saved exact result	Count / outcome
Representative rows / old features	477 / 576
Nonzero old / new terms	476 / 1
Independent identity points	479 passed
Known RHS factor-reuse checks	2 full-field identities passed
Altered programs	2 rejected
Small linear-algebra checks	4 passed

Backend. Ordinary rectangular row elimination retains swaps, inverses, lower multipliers and upper rows for later RHS solves. Free coefficients are zero. The observed producer rank is477; no new independent rank replay is claimed. Global program verification establishes this representation without relying on the old rank premise. Persisted-factor reload has not yet been separately tested.

Costs / custody. Dense exact reference backend: derived factor work O(nmt), RHS work O(nt+mt), plus feature generation and checking. No compressed construction or timing gain is claimed. Sources fixed atc311b8a before the single bounded local invocation: one CPU,512MiB,30seconds,8MiB output cap, exit0. Saved output1,243,148 bytes. No cloud job, build or package install.

Next. Adapt the explicit program to the original point-feature evaluator and decoder. Preserve original recovery cohorts and controls, and admit a complete cold/reuse cost comparison before claiming savings. No unknown-scalar recovery targets were executed in this certificate construction.

Sources. Saved feature-span-dual elimination; feature-fiber-repair packet; NIST DLMF27.9; certificate/program.json, certificate/summary.json and the construction receipt. Program and reusable factor are preserved by SHA256.